

Stat 215A Lab0

/scratch/users/hao_huang/conda/envs/215a/bin/python

1.1 Loading the data

	State	lon	lat
0	Alabama	-86.592392	34.735676
1	Alaska	-168.187471	53.330810
2	Arizona	-111.615136	36.246377
3	Arkansas	-92.240455	36.374502
4	California	-120.386476	34.034180

	State	Murder	Assault	UrbanPop	Rape
0	Alabama	13.2	236	58	21.2
1	Alaska	10.0	263	48	44.5
2	Arizona	8.1	294	80	31.0
3	Arkansas	8.8	190	50	19.5
4	California	9.0	276	91	40.6

1.2 Manipulating the data

```
Index(['State', 'Murder', 'Assault', 'UrbanPop', 'Rape'], dtype='str')
```

```
Index(['State', 'lon', 'lat'], dtype='str')
```

```
Index(['State', 'lon', 'lat'], dtype='str')
```

State	Murder	Assault	UrbanPop	Rape	lon	lat
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```
['Alabama', 'Alaska', 'Arizona', 'Arkansas', 'California']
```

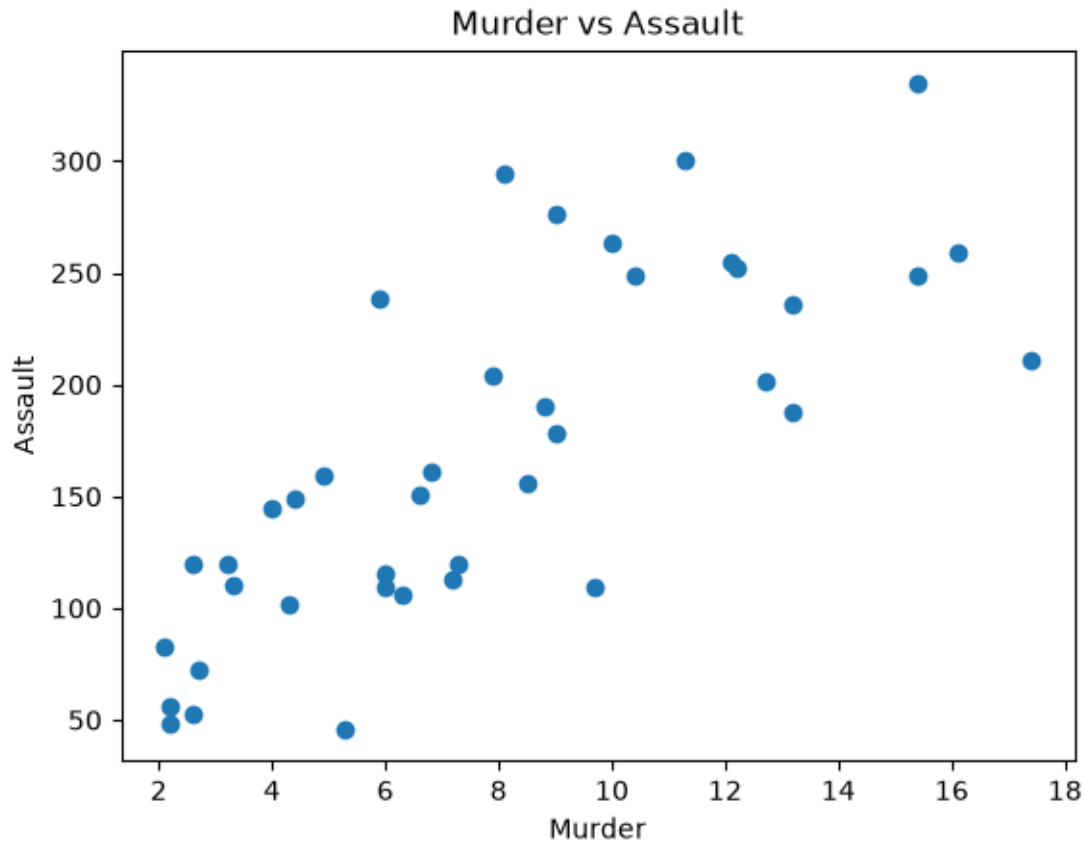
```
['Alabama', 'Alaska', 'Arizona', 'Arkansas', 'California']
```

```
['Alabama', 'Alaska', 'Arizona', 'Arkansas', 'California']
```

	State	Murder	Assault	UrbanPop	Rape	lon	lat
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4	California	9.0	276	91	40.6	-120.386476	34.034180

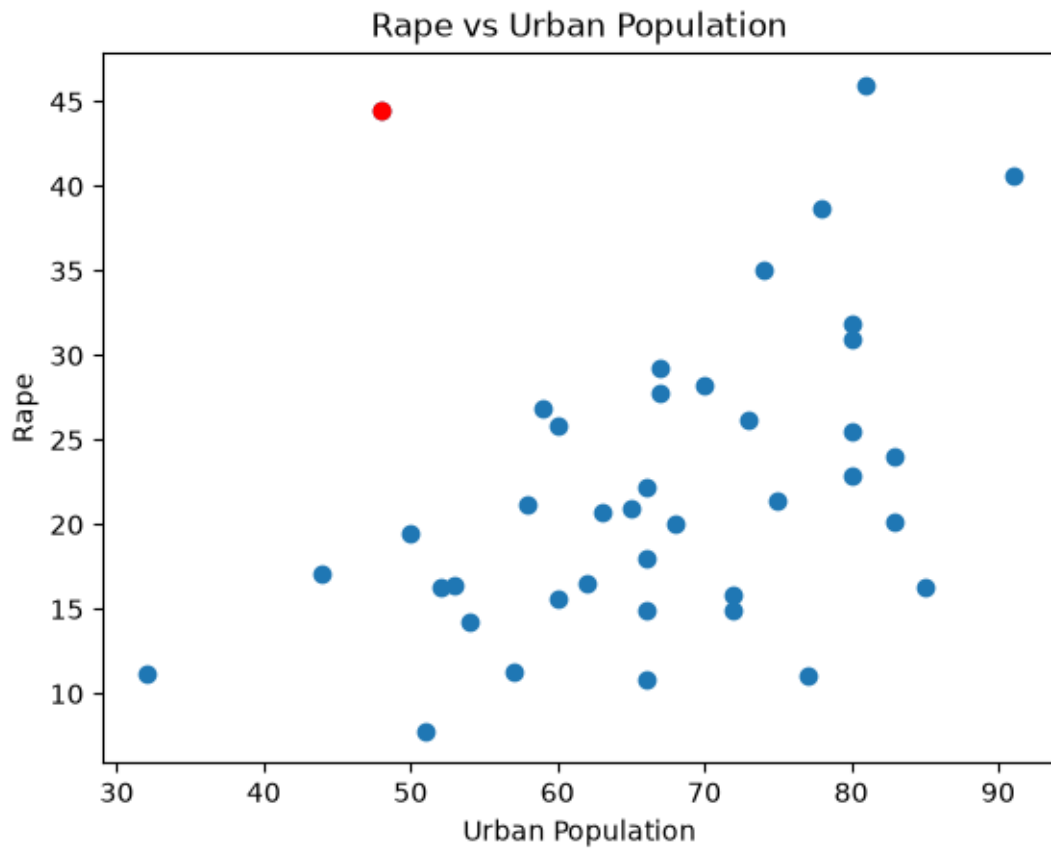
1.3 Visualizaing the data

1) murder vs assault

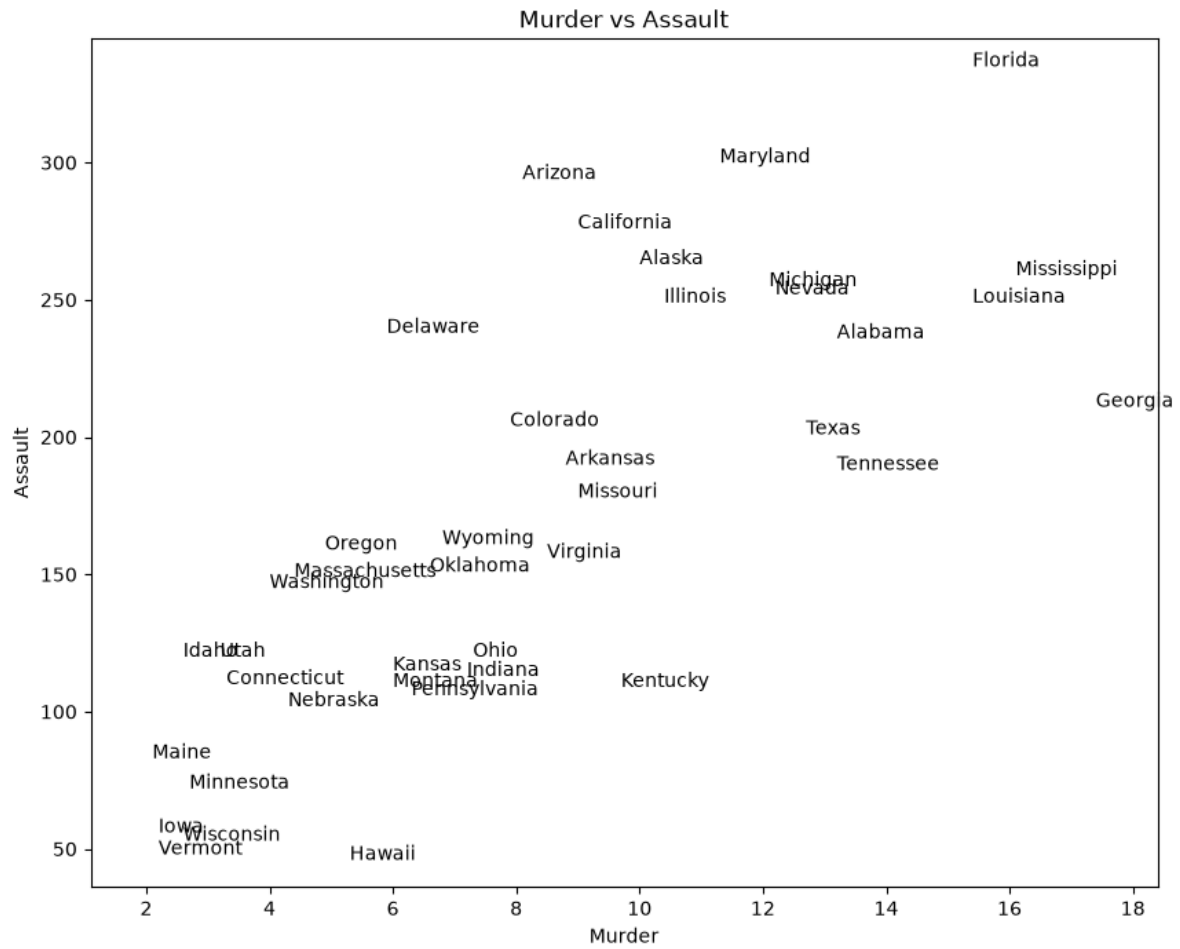


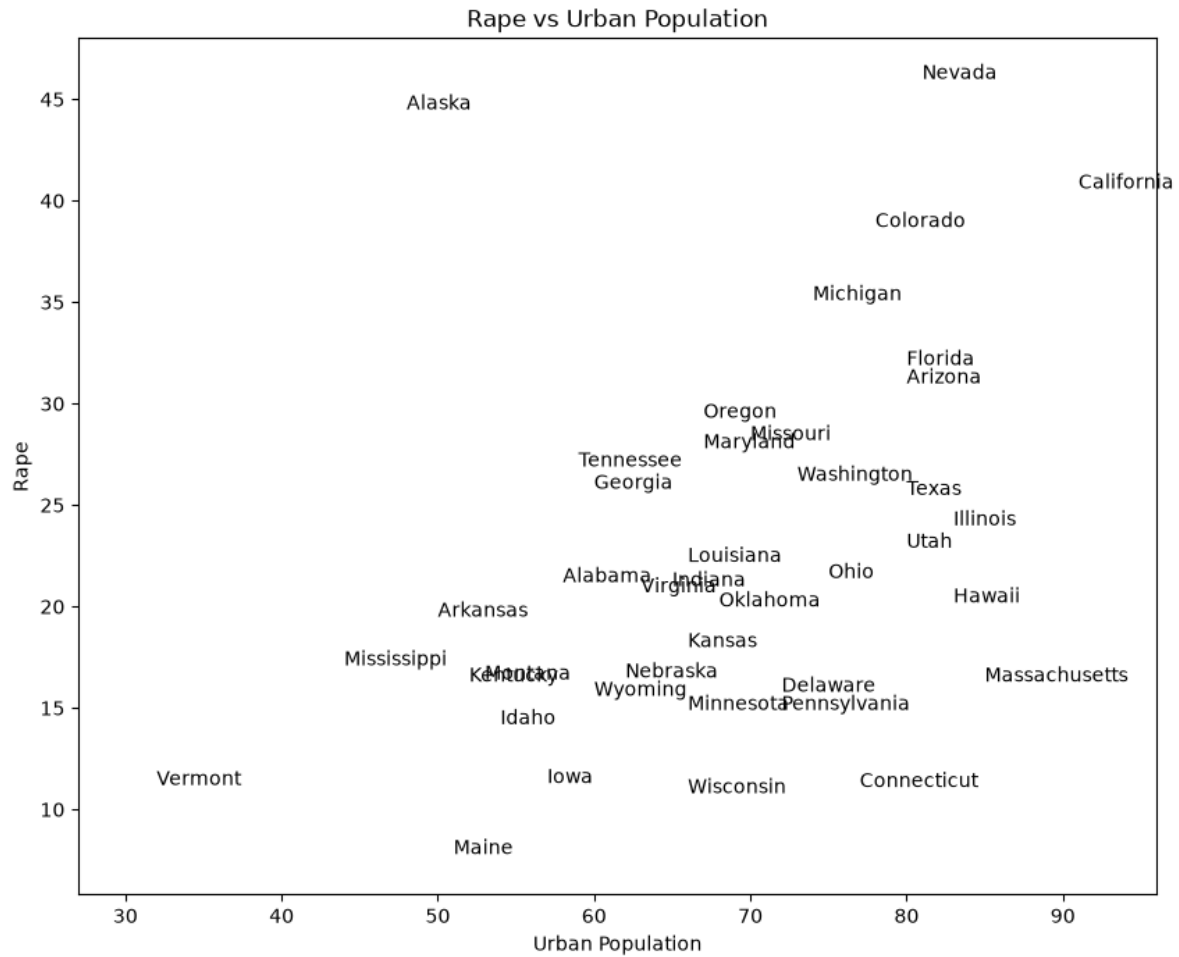
States with higher murder rates also tend to have higher assault rates. Strong positive relationship

2) rape vs urban population



3) remake





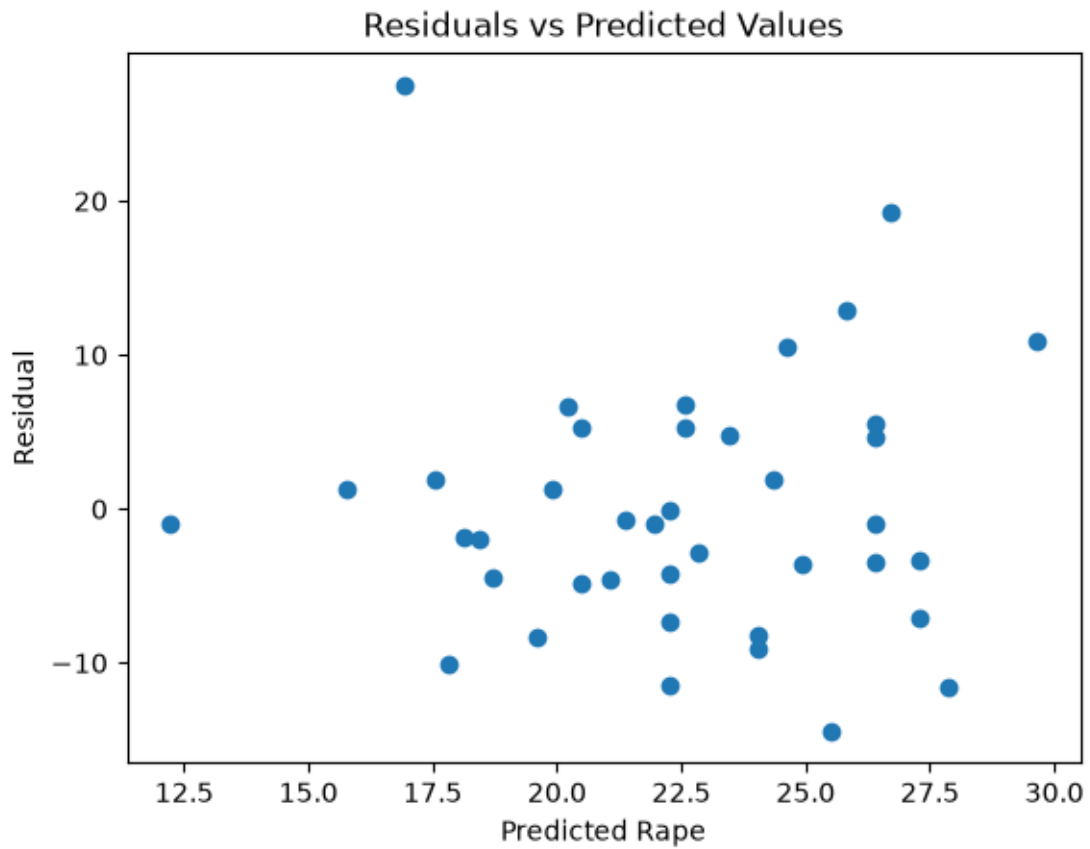
1.4 regression

1)

	State	UrbanPop	Rape	lon	lat
0	Alabama	58	21.2	-86.592392	34.735676
1	Alaska	48	44.5	-168.187471	53.330810
2	Arizona	80	31.0	-111.615136	36.246377
3	Arkansas	50	19.5	-92.240455	36.374502
4	California	91	40.6	-120.386476	34.034180

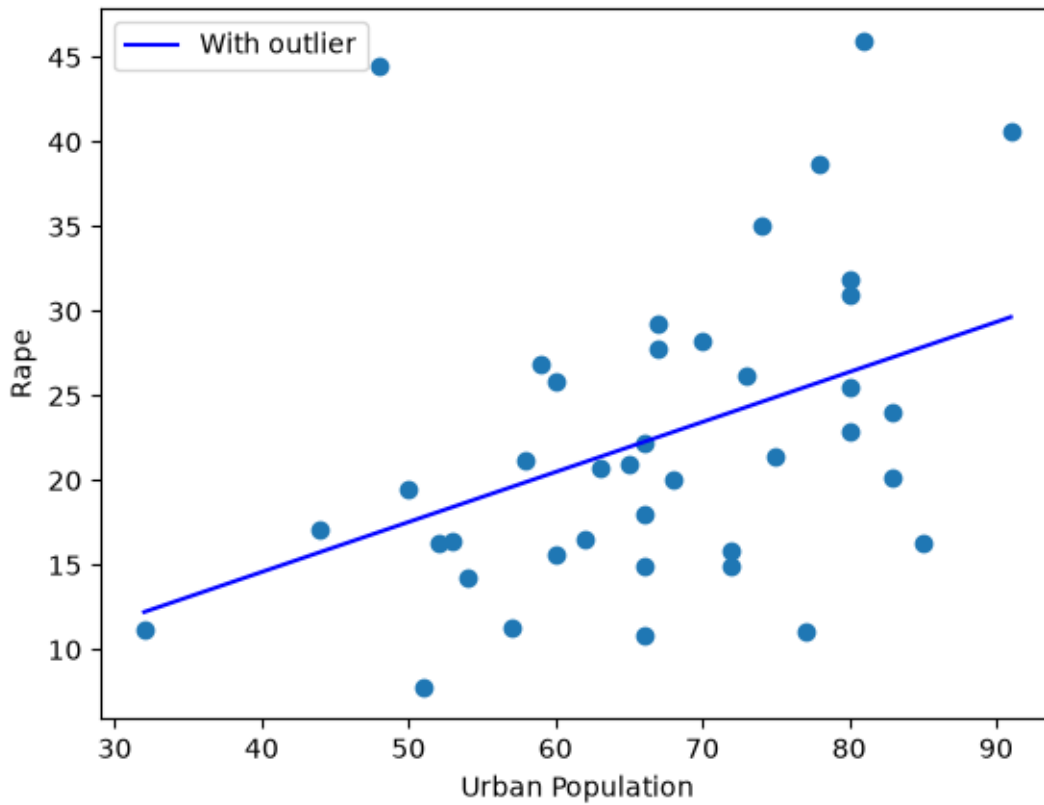
2)

3)



4)

```
/scratch/users/hao_huang/conda/envs/215a/lib/python3.14/site-packages/sklearn/utils/validation.py:100: UserWarning:
warnings.warn(
```



5)

`fit_intercept` `fit_intercept:` True
 bool, default=True

Whether to calculate the intercept for this model. If set to False, no intercept will be used in calculations (i.e. data is expected to be centered).

`copy_X` `copy_X:` bool, True
 default=True

If True, X will be copied; else, it may be overwritten.

tol tol: float, default=1e-6 1e-06

The precision of the solution ('coef_') is determined by 'tol' which specifies the convergence criterion of the underlying solver. 'tol' is set as 'atol' and 'btol' of :func:'scipy.sparse.linalg.lsqr' when fitting on sparse training data. 'tol' is set as 'cond' of :func:'scipy.linalg.lstsq' when fitting on dense training data.

.. versionadded:: 1.7
.. versionchanged:: 1.9
Now supported on dense data, interpreted as the 'cond' parameter.
n_jobs n_jobs: int, None
default=None

The number of jobs to use for the computation. This will only provide speedup in case of sufficiently large problems, that is if firstly 'n_targets > 1' and secondly 'X' is sparse or if 'positive' is set to 'True'. "None" means 1 unless in a :obj:'joblib.parallel_backend' context. "-1" means using all processors. See :term:'Glossary <n_jobs>' for more details.

	positive: bool, default=False	False
	When set to “True“, forces the coefficients to be positive. This option is only supported for dense arrays. For a comparison between a linear regression model with positive constraints on the regression coefficients and a linear regression without such constraints, see :ref:‘sphx_glr_auto_examples_linear_model_plot_nnls.py‘. .. versionadded:: 0.24	

Name	Type	Value
coef_ coef_: array of shape (n_features,) or (n_targets, n_features)	ndarray[float64](1,)	[0.38]
Estimated coefficients for the linear regression problem. If multiple targets are passed during the fit (y 2D), this is a 2D array of shape (n_targets, n_features), while if only one target is passed, this is a 1D array of length n_features.		

feature_names_in_ feature_names_in_: ndarray of shape ('n_features_in_',)	ndarray[object](1,)	['UrbanPop']
Names of features seen during :term:'fit'. Defined only when 'X' has feature names that are all strings.		
.. versionadded:: 1.0 intercept_ intercept_: float or array of shape (n_targets,)	float64	-3.865
Independent term in the linear model. Set to 0.0 if 'fit_intercept = False'. n_features_in_ n_features_in_: int		
	int	1
Number of features seen during :term:'fit'.		
.. versionadded:: 0.24 rank_ rank_: int	int64	np.int64(1)
Rank of matrix 'X'. Only available when 'X' is dense. singular_ singular_: array of shape (min(X, y),)		
	ndarray[float64](1,)	[77.42]
Singular values of 'X'. Only available when 'X' is dense.		
